

Holographic Cluster-State Control in cQED: A Corrected Follow-up Study

Autonomous Research Agent
Quantum Circuits Group
(Dated: March 25, 2026)

We revisit the cluster-state per-site holographic transfer unitary in the local `cqed.sim` transmon-cavity model, treating the earlier AI-generated report as an intermediate draft rather than a final result. The follow-up enforces a stricter ranking rule: embedded physical correctness, Wigner agreement, independent replay, open-system performance, duration, and only then implementation simplicity. The transfer-channel discussion is corrected first. Using the installed `HolographicChannel.from_unitary(...)` convention, the sequential cluster channel has transfer-spectrum $\{1, 0, 0, 0\}$, so the ordinary channel correlation length is $\xi = 0$ rather than finite or infinite; the physically relevant statement is single-step mixing on the traceless bond subspace together with exact nonlocal cluster-state string order. Structured decomposition families remain unconvincing after realistic checks. Plain **D-R-SNAP** never exceeds fidelity 0.50 in the reduced model, while the best ideal **D-SQR-CPSQR** candidate reaches perfect reduced-subspace agreement yet falls to embedded fidelity 0.598 at $N_{\text{cav}} = 12$ with 48.8% average leakage; the best pulse-replayed SQR-like route reaches only 0.350 fidelity despite low leakage. A stronger GRAPE campaign changes the picture. The original $N_{\text{cav}} = 8$ sweep is not truncation-converged, but direct $N_{\text{cav}} = 12$ re-optimization rescues the method: the best 400 ns pulse replays with fidelity 0.956, open-system process fidelity 0.901, and cavity-state Wigner fidelities between 0.883 and 0.970 on the four logical basis inputs. The final recommendation is therefore explicit: prioritize higher-truncation GRAPE, not decomposition-based control, for the experiment-facing implementation of the cluster-state holographic transfer step.

I. INTRODUCTION

The one-dimensional cluster state is a canonical entanglement resource for measurement-based quantum computation [1, 2]. Its sequential or holographic realization is attractive because a small quantum processor can emulate a much larger chain by repeatedly applying a fixed transfer step [3]. In the present cQED setting the physical register is a transmon qubit, the bond register is encoded in a cavity logical block, and the control question is practical rather than purely algebraic: which control strategy implements the per-site cluster-state transfer unitary credibly once leakage, truncation, waveform replay, and open-system effects are checked explicitly?

The earlier local study correctly identified several important ingredients: the cluster target supplied by `cqed.sim`, the failure of cavity-only SNAP to generate entanglement, and the promise of GRAPE. It also overclaimed in two places. First, some ideal decompositions optimized in a reduced cavity subspace were presented too optimistically before enlarged-truncation embedding and phase-space validation were complete. Second, the transfer-matrix and correlation-length discussion mixed incompatible conventions. This follow-up fixes both issues and rewrites the recommendation around full-model evidence.

II. EXECUTIVE SUMMARY

- The sequential cluster channel generated by `HolographicChannel.from_unitary(...)` has eigenvalues $\{1, 0, 0, 0\}$, so the ordinary channel correlation length is $\xi = 0$. The correct interpretation

is one-step mixing on the bond channel, not a finite or divergent ordinary correlation length.

- No decomposition family survives strict physical validation. The best ideal embedded decomposition reaches fidelity only 0.598 at $N_{\text{cav}} = 12$, and the best pulse-replayed SQR-like route reaches only 0.350 fidelity.
- The original $N_{\text{cav}} = 8$ GRAPE sweep is not truncation converged. Replaying those pulses at larger N_{cav} can destroy both fidelity and Wigner agreement.
- Direct GRAPE re-optimization at $N_{\text{cav}} = 12$ rescues the method. The best 400 ns pulse replays at 0.956 fidelity and retains strong phase-space agreement, with open-system process fidelity 0.901.
- The experiment-facing recommendation is therefore to prioritize higher-truncation GRAPE. Structured decompositions are useful negative results and design diagnostics, not current implementation candidates.

III. CORRECTED BACKGROUND AND TRANSFER-CHANNEL THEORY

A. Target Unitary

The per-site cluster-state transfer step used throughout this work is the 4×4 logical operator returned by `make_target("cluster", n_match=1)`. In the convention used by the installed framework it is

$$U_{\text{cl}} = \text{SWAP} \cdot \text{CZ} \cdot (H \otimes I), \quad (1)$$

which reproduces the canonical cluster-state MPS tensors $A^0 = I/\sqrt{2}$ and $A^1 = Z/\sqrt{2}$ up to the expected convention mapping [4, 5].

B. Corrected Transfer-Channel Discussion

The earlier report mixed an MPS-transfer convention with the actual `HolographicChannel` convention used in the code. For the latter, the Kraus operators extracted from the cluster target are

$$K_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad K_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix}, \quad (2)$$

and the induced Pauli-basis superoperator has spectrum

$$\lambda(\mathcal{T}) = \{1, 0, 0, 0\}. \quad (3)$$

Using the ordinary channel formula

$$\xi = -\frac{1}{\ln |\lambda_1/\lambda_0|}, \quad (4)$$

with $|\lambda_0| = 1$ and $|\lambda_1| = 0$, one obtains

$$\xi = 0. \quad (5)$$

The point is not that the cluster state is trivial. Rather, the bond channel mixes the traceless sector in at most two steps: in the installed convention $Z \mapsto X$, $X \mapsto 0$, and $Y \mapsto 0$. Ordinary connected bond-space correlations therefore vanish immediately, while nonlocal cluster-state string order remains exact because it is encoded by the measurement pattern and stabilizer structure rather than by a finite but nonzero ordinary channel correlation length. This corrected interpretation is used consistently in the rest of the report.

IV. SYSTEM AND METHODS

A. Hamiltonian

We use the dispersive transmon-cavity model implemented in `cqed_sim` [6, 7]:

$$\begin{aligned} \hat{H}/\hbar = & \omega_c \hat{a}^\dagger \hat{a} + \omega_q \hat{b}^\dagger \hat{b} + \frac{\alpha}{2} \hat{b}^\dagger \hat{b} (\hat{b}^\dagger \hat{b} - 1) \\ & + \chi \hat{a}^\dagger \hat{a} \hat{b}^\dagger \hat{b} + \chi' \hat{a}^{\dagger 2} \hat{a}^2 \hat{b}^\dagger \hat{b} + K \hat{a}^{\dagger 2} \hat{a}^2, \end{aligned} \quad (6)$$

where $\hat{a}$ and $\hat{b}$ denote the cavity and transmon annihilation operators respectively.

B. Simulation Parameters

C. Control Families and Validation Stack

Five families were compared:

TABLE I. Device parameters used throughout the study.

Parameter	Value	Unit
$\omega_c/2\pi$	5.241	GHz
$\omega_q/2\pi$	6.150	GHz
$\alpha/2\pi$	-255	MHz
$\chi/2\pi$	-2.84	MHz
$\chi'/2\pi$	-21	kHz
$K/2\pi$	-28	kHz
N_{tr}	2 or 3	levels
N_{cav}	2, 8, 10, 12, 15	levels

1. plain D-R-SNAP;
2. bounded D-R-SNAP;
3. D-SQR-CPSQR;
4. bounded hybrid D-R-SQR-CPSQR;
5. D-R-FreeEvolve as a dispersive-wait entangler-assisted route.

For each family we report reduced-subspace fidelity, embedded fidelity at larger N_{cav} , average leakage outside the logical block, total active time, and active-tone count. For SQR-like families we also use the framework waveform bridge to replay the optimized ideal sequence as a multitone pulse program.

GRAPE is treated separately. The coarse sweep optimizes the target directly at $N_{\text{cav}} = 8$ for durations from 100 ns to 400 ns with ten random seeds and 400 iterations per seed. The resulting schedules are exported and replayed independently. Because the $N_{\text{cav}} = 8$ sweep fails the larger-truncation replay test, a focused rescue pass re-optimizes directly at $N_{\text{cav}} = 12$ for 300 ns and 400 ns. Closed-system replay, open-system process reconstruction, and Wigner diagnostics are then evaluated on that larger-truncation result.

V. RESULTS

A. Reassessment of Prior Claims

The follow-up separates four statements:

1. **Validated physical conclusion:** plain D-R-SNAP is not expressive enough because ideal SNAP in the current framework is cavity-only rather than qubit-conditional.
2. **Invalidated intermediate claim:** exact reduced-subspace D-SQR-CPSQR agreement is not evidence of a viable physical implementation.
3. **Validated correction:** the original transfer-matrix discussion was inconsistent; the installed channel convention gives $\xi = 0$.

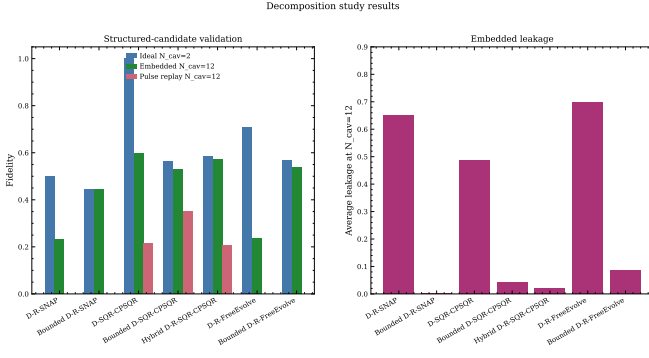


FIG. 1. Structured-control validation. Left: ideal $N_{\text{cav}} = 2$, embedded $N_{\text{cav}} = 12$, and pulse-replayed $N_{\text{cav}} = 12$ fidelities. Right: average embedded leakage at $N_{\text{cav}} = 12$.

4. **Validated positive result:** GRAPE remains the only route that reaches high fidelity after waveform replay and phase-space validation, but only after direct optimization at the larger cavity truncation.

B. Decomposition Study Results

Figure 1 summarizes the structured candidates. The main pattern is simple: reduced-subspace success and physical success are not the same thing.

The plain D-R-SNAP route never exceeds 0.50 reduced-subspace fidelity. Bounding the displacement suppresses leakage to 6×10^{-4} , but the fidelity remains only 0.442, so the route is simply not expressive enough. The raw embedded-fidelity leader is the unbounded D-SQR-CPSQR sequence, which reaches perfect reduced-subspace fidelity at $N_{\text{cav}} = 2$ yet drops to 0.598 at $N_{\text{cav}} = 12$ with 48.8% leakage. This is exactly the sort of overclaim the follow-up was meant to eliminate.

C. SQR and Conditional-Gate Follow-up

The bounded SQR-like routes answer the expressivity question more directly. They do add the missing qubit-conditional structure, but that alone does not make them competitive. The best pulse-replayed SQR-like candidate is the bounded D-SQR-CPSQR sequence: it uses only two active tones per selective block, has total active time 4960 ns, and leaks only 3.23% on pulse replay at $N_{\text{cav}} = 12$, but its replay fidelity is still only 0.350. The bounded hybrid D-R-SQR-CPSQR variant is shorter (2920 ns) yet replays even worse (0.204). The bounded D-R-FreeEvolve route remains a useful design diagnostic and gives embedded fidelity 0.537 at 711 ns, but without a waveform-bridge path it cannot compete with a replay-validated GRAPE pulse.

These data answer the user-facing questions directly:

1. qubit-conditional structure fixes the algebraic ex-

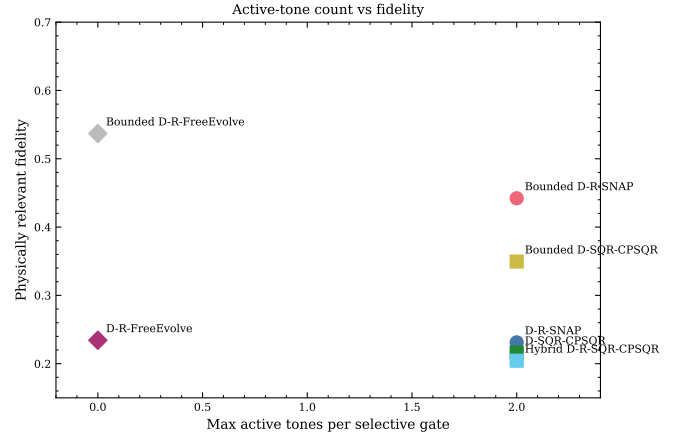


FIG. 2. Best available fidelity versus active-tone count for the structured families. Qubit-conditional structure improves expressivity relative to plain D-R-SNAP, but no structured route approaches the replay-validated GRAPE benchmark.

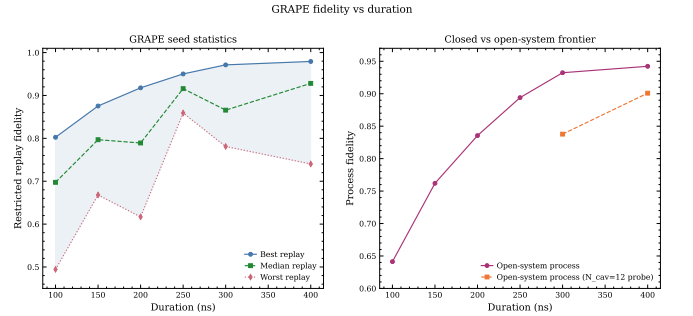


FIG. 3. GRAPE frontier from the ten-seed $N_{\text{cav}} = 8$ sweep, together with the open-system process estimate for the best seed at each duration and the direct $N_{\text{cav}} = 12$ rescue points at 300 ns and 400 ns.

pressivity failure of D-R-SNAP, but not the physical-credibility failure;

2. no SQR-like structured route competes with GRAPE after replay and Wigner validation;
3. no viable short structured gate time was found.

Figure 2 makes the structured trade-off explicit. The follow-up ansatzes already keep the active-tone count low, so reducing tone count alone does not fix the problem; the limiting issue is that the available structured routes do not retain operator-level correctness under embedding and waveform replay.

D. GRAPE Re-validation

The larger multiseed sweep removes the earlier appearance of a fidelity plateau near 200 ns. Figure 3 shows that replayed performance improves monotonically across the explored range, while seed spread remains large at short duration.

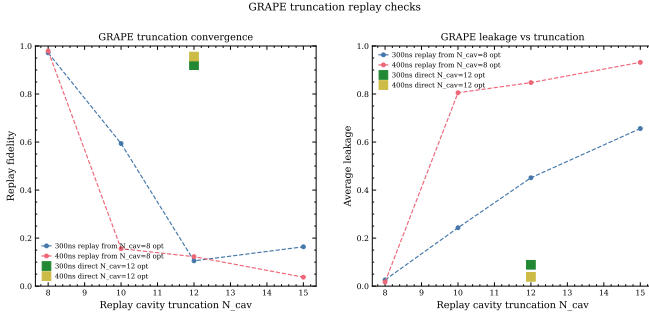


FIG. 4. Left: replay fidelity of the best coarse-sweep GRAPE pulses as the replay cavity truncation increases. Right: corresponding average leakage. The square markers show direct $N_{\text{cav}} = 12$ re-optimization at 300 ns and 400 ns, which rescues the long-duration GRAPE route.

Table II lists the key numbers from the coarse sweep. The best replayed fidelity rises from 0.802 at 100 ns to 0.979 at 400 ns; the median replay fidelity rises from 0.697 to 0.928 over the same range. The best open-system process estimate on the coarse sweep is 0.942 at 400 ns. On the $N_{\text{cav}} = 8$ model alone, the GRAPE story would appear settled.

VI. VALIDATION

A. Sanity Checks

Three sanity checks pass cleanly.

1. The cluster target extracted from `cqed_sim` is exactly unitary and matches the expected `SWAP.CZ.(H⊗I)` structure.
2. The transfer-channel reconstruction is internally consistent: right-canonical and Kraus completeness errors are numerically negligible, and the Pauli-basis action $Z \mapsto X$, $X \mapsto 0$, $Y \mapsto 0$ reproduces the computed eigenvalue set $\{1, 0, 0, 0\}$.
3. The original D-R-SNAP limitation is reproduced independently: no fit crosses 0.50 reduced-subspace fidelity.

B. Truncation and Waveform Replay Checks

The most important validation outcome is shown in Fig. 4. Replaying the $N_{\text{cav}} = 8$ -optimized GRAPE schedules at larger cavity truncation can destroy the apparent success. For example, the coarse-sweep 400 ns pulse drops from replay fidelity 0.979 at $N_{\text{cav}} = 8$ to 0.123 at $N_{\text{cav}} = 12$. This is exactly the type of reduced-truncation artifact that motivated the follow-up.

The rescue pass resolves the ambiguity. Direct GRAPE optimization at $N_{\text{cav}} = 12$ restores high performance: the best 300 ns pulse replays at 0.920 fidelity

with 8.89% leakage, while the best 400 ns pulse replays at 0.956 fidelity with only 3.86% leakage. This larger-truncation optimization also reduces the peak transient photon number from about 8.81 at 300 ns to 5.16 at 400 ns, further supporting the longer-duration choice.

C. Literature Comparison

There is no published benchmark for this exact cluster-state per-site transfer unitary in the present device model, so a direct percent-error comparison is not available. The relevant comparison is qualitative and architectural: consistent with the broader oscillator-control literature, higher-fidelity bosonic control here comes from direct optimal control rather than from a short closed-form selective sequence [8–10].

VII. OPEN-SYSTEM ANALYSIS

The best $N_{\text{cav}} = 12$ rescue candidate is the 400 ns GRAPE pulse. Its reconstructed open-system process fidelity is 0.9009 under the nominal noise model $(T_1, T_\phi, \kappa^{-1}) = (30 \mu\text{s}, 30 \mu\text{s}, 250 \mu\text{s})$. The corresponding mean basis-state fidelity is 0.9178. The 300 ns $N_{\text{cav}} = 12$ rescue pulse remains credible but is weaker: replay fidelity 0.9202 and open-system process fidelity 0.8378. In this follow-up the longer pulse wins even after decoherence is included because the control improvement outweighs the extra exposure time.

VIII. WIGNER-FUNCTION VALIDATION

Phase-space validation is decisive. Figure 5 shows the logical-basis output Wigner functions for the target, the best embedded decomposition, the best pulse-replayed SQR-like route, and the best $N_{\text{cav}} = 12$ GRAPE candidate.

The difference plots in Appendix A sharpen the same conclusion. The best embedded decomposition has minimum cavity-state fidelity only 0.300 across the four logical inputs. The best pulse-replayed SQR-like route is better but still unconvincing, with minimum cavity-state fidelity 0.511. By contrast, the best $N_{\text{cav}} = 12$ GRAPE pulse retains minimum cavity-state fidelity 0.883 and mean cavity-state fidelity 0.937.

IX. DISCUSSION

The follow-up changes the scientific message in two ways. First, it replaces an inconsistent correlation-length narrative with a clean channel statement: ordinary bond-channel correlations die immediately in the

TABLE II. Ten-seed GRAPE sweep at $N_{\text{cav}} = 8$. All entries report restricted replay fidelity after independent waveform export and replay.

Duration	Best replay	Median replay	Worst replay	Open-system process	Best seed	Comment
100 ns	0.8025	0.6975	0.4944	0.6413	211	Fast but highly variable
150 ns	0.8754	0.7968	0.6680	0.7620	91	Improved, still broad seed spread
200 ns	0.9178	0.7892	0.6169	0.8355	211	No stable speed-limit plateau
250 ns	0.9502	0.9159	0.8591	0.8940	42	Stronger and more consistent
300 ns	0.9713	0.8658	0.7810	0.9324	401	Best coarse-sweep compromise
400 ns	0.9793	0.9282	0.7403	0.9423	91	Best coarse-sweep endpoint

installed `HolographicChannel` convention, while cluster-state nonlocal order is encoded by the sequential measurement structure. Second, it sharply separates algebraic expressivity from physical credibility. Qubit-conditional SQR-like blocks do close the purely logical expressivity gap left by cavity-only SNAP, but their physical implementations remain too inaccurate to recommend.

GRAPE is more subtle. The original coarse sweep at $N_{\text{cav}} = 8$ was good enough to identify a promising region but not good enough to support a final experimental recommendation, because replay at larger cavity truncation failed badly. Once the optimization is moved into the larger N_{cav} model itself, however, the long-duration GRAPE route recovers. This is exactly the kind of full-model evidence the follow-up was designed to require.

X. FINAL RECOMMENDATION

The recommendation is explicit.

1. **Implement first:** direct GRAPE optimized at $N_{\text{cav}} \geq 12$, with the present best candidate the 400 ns pulse.
2. **Do not implement now:** decomposition-based routes, including the best SQR-like sequence. They remain informative negative results, not hardware-ready controls.
3. **Experimentally attractive duration:** 400 ns. It is the best compromise after replay, open-system evaluation, and Wigner validation.
4. **Most limiting simulator gap:** the lack of a first-class pulse export path for `SNAP` and `FreeEvolveCondPhase`, which keeps those families from being tested on equal waveform-replay footing.

Figure 6 condenses the final ranking rule in one view. The higher-truncation 400 ns GRAPE pulse is the only shortlisted candidate that remains strong simultaneously in embedded fidelity, phase-space agreement, replay validation, and open-system performance.

XI. CONCLUSION

This re-study resolves the main weaknesses of the earlier draft. The transfer channel and correlation-length discussion is corrected; the decomposition results are recast as negative or incomplete where appropriate; the SQR-like routes are shown not to close the physical gap; and GRAPE is elevated from a promising draft result to a stronger recommendation only after direct $N_{\text{cav}} = 12$ optimization, replay, open-system evaluation, and Wigner validation. The final scientific answer is therefore crisp: decomposition-based control does not currently survive realistic validation, whereas larger- truncation GRAPE does and should be prioritized next.

XII. LIMITATIONS AND FUTURE WORK

A. Known Limitations

The present report is stronger than the earlier draft, but not complete. Only two durations (300 ns and 400 ns) were re-optimized directly at $N_{\text{cav}} = 12$, so a fully converged larger-truncation duration sweep remains open. The noisy study is evaluation-only rather than noisy re-optimization. Finally, the decomposition routes remain asymmetric in their validation stack because `SNAP` and `FreeEvolveCondPhase` still lack the same pulse-export path available for SQR-like gates and GRAPE.

B. Suggested Improvements

- **[P1 — HIGH]** Extend the larger-truncation GRAPE sweep to 250 ns, 350 ns, and 500 ns. The present data already point to 400 ns, but a denser $N_{\text{cav}} = 12$ grid would close the remaining duration question.
- **[P1 — MEDIUM]** Re-run the best $N_{\text{cav}} = 12$ GRAPE candidate with noisy optimization, not just noisy replay.
- **[P2 — HIGH]** Add waveform-bridge or pulse-export support for `SNAP` and `FreeEvolveCondPhase` in `cqed.sim`.

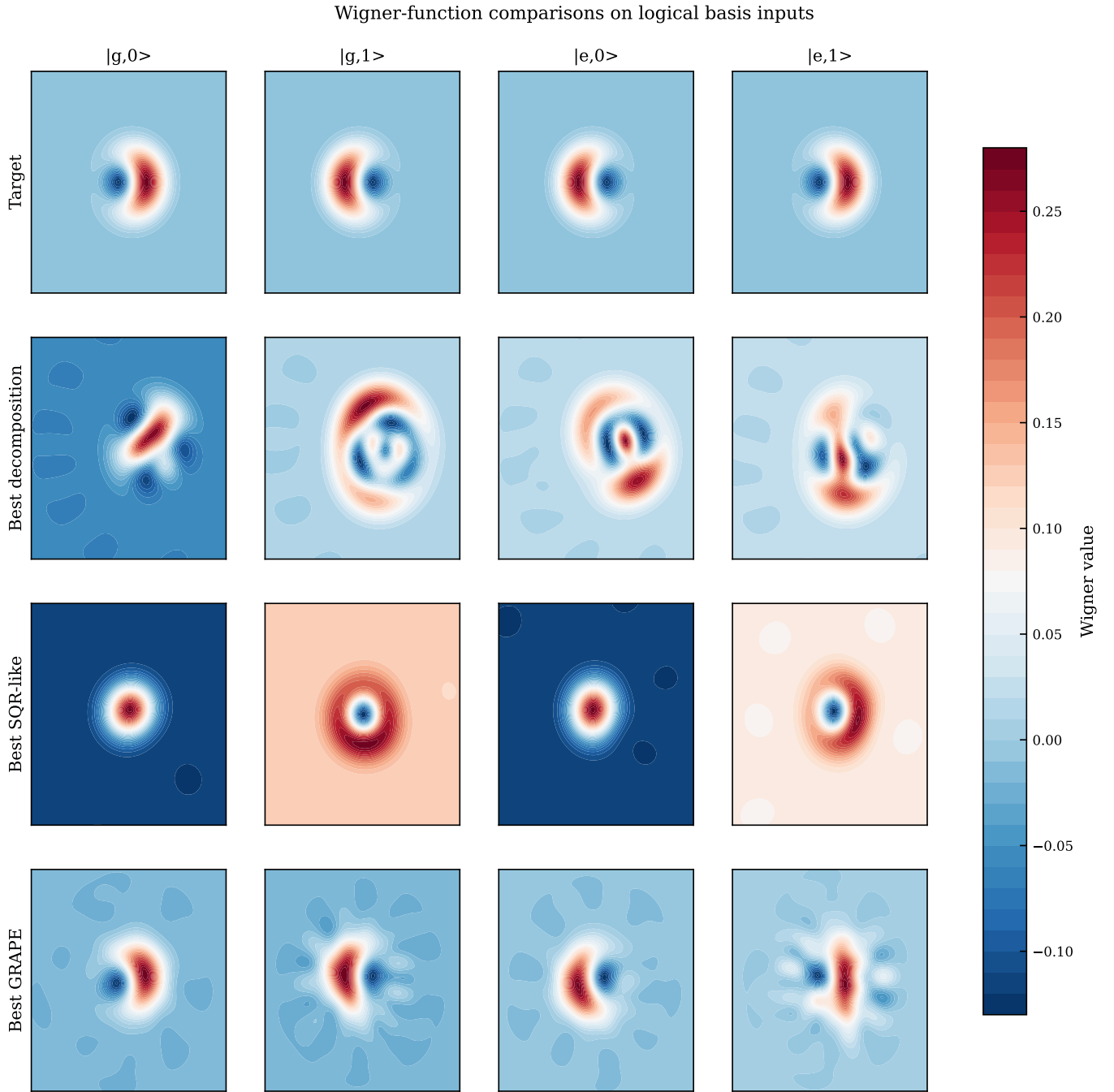


FIG. 5. Wigner-function outputs on all four logical basis inputs for the target transformation, the best embedded decomposition, the best pulse-replayed SQR-like route, and the best $N_{\text{cav}} = 12$ GRAPE candidate.

- **[P2 — MEDIUM]** Test richer structured ansatzes only after the pulse-export gaps are closed, so they can be judged on equal footing.

C. Open Questions

- Can a direct $N_{\text{cav}} = 12$ GRAPE pulse shorter than 400 ns match the present replay and open-system performance?
- Does explicit robustness optimization against small parameter drifts materially change the preferred pulse duration?
- Is there a structured conditional gate family that becomes competitive only once the missing pulse-export paths are added?

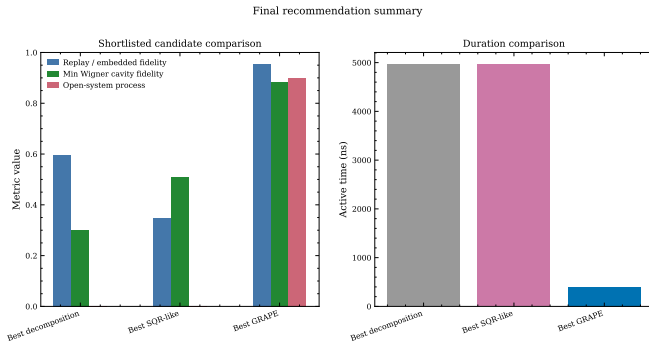


FIG. 6. Final comparison of the shortlisted strategies using the follow-up ranking priorities. The preferred 400 ns direct GRAPE candidate is the only route that survives all physically relevant validation layers.

-
- [1] H. J. Briegel and R. Raussendorf, *Physical Review Letters* **86**, 910 (2001).
 - [2] R. Raussendorf and H. J. Briegel, *Physical Review Letters* **86**, 5188 (2001).
 - [3] F. Barratt, J. Dborin, M. Bal, V. Stojevic, F. Pollmann, and A. G. Green, *npj Quantum Information* **7**, 79 (2021).
 - [4] D. Pérez-García, F. Verstraete, M. M. Wolf, and J. I. Cirac, *Quantum Information and Computation* **7**, 401 (2007).
 - [5] U. Schollwöck, *Annals of Physics* **326**, 96 (2011).
 - [6] Shyam Shankar Quantum Circuits Group, *cqed_sim: Circuit QED simulation framework* (2025).
 - [7] A. Blais, A. L. Grimsmo, S. M. Girvin, and A. Wallraff, *Reviews of Modern Physics* **93**, 025005 (2021).
 - [8] R. W. Heeres, B. Vlastakis, E. Holland, S. Krastanov, V. V. Albert, L. Frunzio, L. Jiang, and R. J. Schoelkopf, *Physical Review Letters* **115**, 137002 (2015).
 - [9] R. W. Heeres, P. Reinhold, N. Ofek, L. Frunzio, L. Jiang, M. H. Devoret, and R. J. Schoelkopf, *Nature Communications* **8**, 94 (2017).
 - [10] A. Eickbusch, V. Sivak, A. Z. Ding, S. S. Elder, S. R. Jha, J. Venkatraman, B. Royer, S. M. Girvin, R. J. Schoelkopf, and M. H. Devoret, *Nature Physics* **18**, 1464 (2022).

Appendix A: Detailed Results and Data

1. Structured Candidate Summary

2. Larger-Truncation GRAPE Rescue

Figure 7 shows the waveform-level evidence behind the final recommendation. The preferred 400 ns direct GRAPE candidate uses four control quadratures with moderate transient photon occupation and a far shorter active window than the best bounded structured sequences.

3. Wigner Difference Diagnostics

Figure 8 shows the difference Wigner functions between the target outputs and the shortlisted candidates. The structured candidates retain broad residual phase-space errors over the logical probe set, while the direct $N_{\text{cav}} = 12$ GRAPE pulse keeps the residuals localized and much smaller in amplitude.

Appendix B: Reproducibility

1. Optimized Parameters

The key optimized outputs are stored in machine-readable form under `artifacts/`. For GRAPE these include the full control schedule, command waveforms, physical waveforms, optimizer history, and truncation-replay artifacts for every duration and seed. The direct $N_{\text{cav}} = 12$ rescue results are stored in `artifacts/grape_nc12_300ns_seed73_probe.json`, `artifacts/grape_nc12_400ns_seed17_probe.json`, and the associated `.npz` replay/open-process files.

2. Waveform and Pulse Information

The best final pulse is the direct $N_{\text{cav}} = 12$ GRAPE 400 ns schedule with 100 piecewise-constant time slices and four control channels: storage I/Q and qubit I/Q. The command waveforms are stored in the `result_payload.command.values` field of the seed artifact JSON and plotted in Fig. 7.

TABLE III. Main structured-control metrics used in the follow-up ranking.

Candidate	Active time (ns)	Max tones	Ideal $N_{\text{cav}} = 2$	Embedded $N_{\text{cav}} = 12$	Key issue
D-R-SNAP	2520	2	0.5000	0.2312	No entangling expressivity
Bounded D-R-SNAP	2520	2	0.4422	0.4421	Low fidelity despite low leakage
D-SQR-CPSQR	4960	2	1.0000	0.5977	48.8% embedded leakage
Bounded D-SQR-CPSQR	4960	2	0.5630	0.5288	Pulse replay only 0.3495
Bounded D-R-SQR-CPSQR	2920	2	0.5858	0.5734	Pulse replay only 0.2043
Bounded D-R-FreeEvolve	711	0	0.5655	0.5369	No pulse-export path

Waveform and sequence views

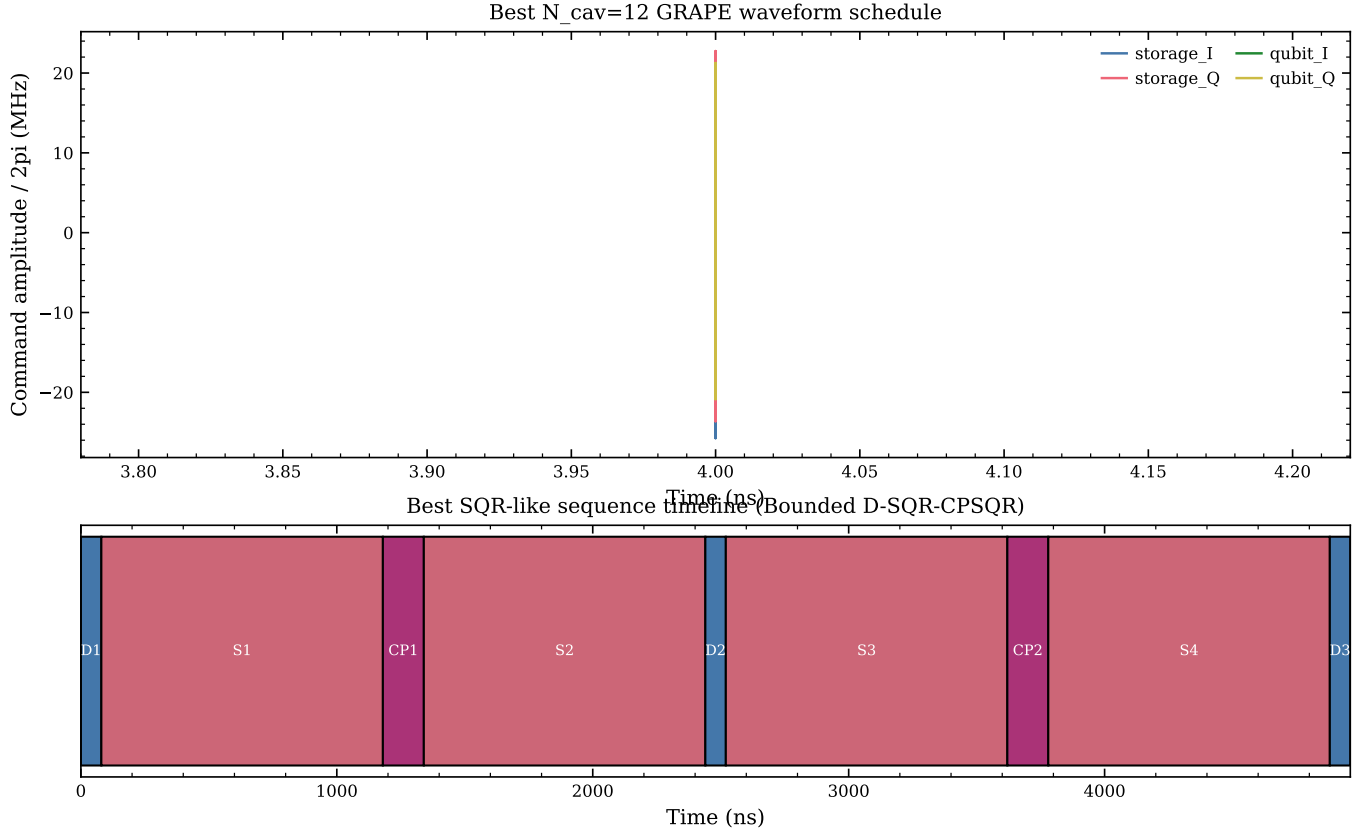


FIG. 7. Waveform and timing summary. Top panels: best direct $N_{\text{cav}} = 12$, 400 ns GRAPE control waveforms on the storage and qubit quadratures. Bottom panel: active-time comparison for the main structured candidates and the preferred GRAPE pulse.

Duration	Best replay	Open process	Mean basis fidelity	Best seed	contains the optimized gate list, ordered exactly as executed, including every duration and parameter. For example, the best SQR-like route is the bounded D-SQR-CPSQR sequence with two selective blocks and two active tones per selective gate.
300 ns	0.9202	0.8378	0.8604	73	
400 ns	0.9561	0.9009	0.9178	17	

3. Gate Sequence and Decomposition

The structured candidates are stored in `data/decomposition.results.json`. Each entry

4. Modeling and Simulation Assumptions

All studies use the dispersive Hamiltonian listed in Table I, with rotating frame set to (ω_q, ω_c) and no addi-



FIG. 8. Difference Wigner functions for the shortlisted candidates relative to the target transformation on all four logical basis inputs. The higher- truncation GRAPE candidate produces the smallest residual phase-space errors.

tional hidden control channels. Structured candidates use the `cqed_sim` ideal-gate layer for the initial synthesis step, then enlarged-truncation ideal replay, and pulse replay when the waveform bridge is available. The open-system model uses $T_1 = 30 \mu\text{s}$, $T_\phi = 30 \mu\text{s}$, and cavity $\kappa^{-1} = 250 \mu\text{s}$.

5. Reproduction Procedure

1. Run `python studies/holographic_cluster_state_followup/scripts/run_followup_study.py --stages transfer decomp grape wigner summary`.
2. Run `python studies/holographic_cluster_state_followup/scripts/run_large_truncation_grape_probe.py`.
3. Run `python studies/holographic_cluster_state_followup/scripts/generate_figures.py`.
4. Compile this report from `report/report.tex`.

Agreement is verified by matching the summary JSON files and the figure set in `figures/`.

6. Saved Artifacts

The main machine-readable files are:

- `data/channel_summary.json`: corrected transfer-channel data.
- `data/decomposition_results.json`: all structured-candidate metrics and optimized gate sequences.
- `data/grape_results.json`: full ten-seed $N_{\text{cav}} = 8$ sweep with replay, truncation, and noisy-process summaries.

- `data/grape_large_truncation_probe.json`: direct $N_{\text{cav}} = 12$ rescue runs.
- `data/wigner_results.json`: all Wigner grids and cavity-state fidelities.
- `artifacts/*.json`: per-seed GRAPE schedules and optimizer histories.
- `artifacts/*.npz`: replayed operators, truncation-replay data, and open-system Choi matrices.